Differentiation & Partial Differentiation

Simple explanation, important rules, step-by-step solved examples, comparison and practice

Learning goal: Understand what a derivative means, how to differentiate common functions, and how partial derivatives work when a function contains more than one variable.

1. What is Differentiation?

Differentiation tells us the **rate of change** of one quantity with respect to another. Geometrically, a derivative gives the **slope of the tangent** to a curve at a particular point.

If y = f(x), the derivative can be written as:

dy/dx or **f'(x)**

Example idea: If distance changes with time, the derivative of distance with respect to time gives velocity.

2. Basic Differentiation Rules

Rule	Function	Derivative
Constant	y = c	dy/dx = 0
Power rule	y = x^n	dy/dx = n x^(n-1)
Constant multiple	y = c f(x)	dy/dx = c f'(x)
Sum rule	y = f(x) + g(x)	dy/dx = f'(x) + g'(x)
Product rule	y = u v	dy/dx = u(dv/dx) + v(du/dx)
Quotient rule	y = u/v	dy/dx = [v(du/dx)-u(dv/dx)]/v^2
Chain rule	y = f(g(x))	dy/dx = f'(g(x)) g'(x)

3. Differentiation - Solved Examples

Example 1: Power rule

Given: y = x^5

Answer: dy/dx = 5x^4

Example 2: Polynomial

Given: y = 3x^2 + 2x + 5

Answer: dy/dx = 6x + 2

Example 3: Cubic polynomial

Given: y = 4x^3 - 2x^2 + 7x - 9

Answer: dy/dx = 12x^2 - 4x + 7

Example 4: Negative power

Given: y = 1/x = x^(-1)

Answer: dy/dx = -x^(-2) = -1/x^2

Example 5: Square root

Given: y = sqrt(x) = x^(1/2)

Answer: dy/dx = (1/2)x^(-1/2) = 1/(2sqrt(x))

Example 6: Product rule

Given: y = x^2(x + 3)

Answer: dy/dx = x^2(1) + (x+3)(2x) = 3x^2 + 6x

Example 7: Chain rule

Given: y = (x^2 + 1)^3

Answer: dy/dx = 3(x^2+1)^2(2x) = 6x(x^2+1)^2

4. What is Partial Differentiation?

Partial differentiation is used when a function depends on **two or more independent variables**. We differentiate with respect to one variable while treating all other variables as constants.

> For z = f(x,y): **partial f / partial x** means differentiate with respect to x while keeping y constant.
> **partial f / partial y** means differentiate with respect to y while keeping x constant.

5. Partial Derivatives - Step-by-Step Examples

Example 1

Given: f(x,y) = x^2 + y^2

With respect to x: y^2 is constant, so partial f/partial x = 2x.

With respect to y: x^2 is constant, so partial f/partial y = 2y.

Example 2

Given: f(x,y) = 3x^2 + 2xy + y^2

With respect to x: 3x^2 -> 6x, 2xy -> 2y, y^2 -> 0.

partial f/partial x = 6x + 2y

With respect to y: 3x^2 -> 0, 2xy -> 2x, y^2 -> 2y.

partial f/partial y = 2x + 2y

Example 3

Given: f(x,y) = x^2 y

With respect to x, treat y as constant: **partial f/partial x = 2xy**.

With respect to y, treat x^2 as constant: **partial f/partial y = x^2**.

Example 4

Given: f(x,y) = x^3 y^2

partial f/partial x = 3x^2 y^2

partial f/partial y = 2x^3 y

Example 5

Given: f(x,y,z) = x^2 + 3xy + yz^2

partial f/partial x = 2x + 3y

partial f/partial y = 3x + z^2

partial f/partial z = 2yz

6. Ordinary vs Partial Differentiation

Feature	Ordinary Differentiation	Partial Differentiation
Typical function	y = f(x)	z = f(x,y) or f(x,y,z)
Variables	Usually one independent variable	Two or more independent variables
Notation	dy/dx, f'(x)	partial f/partial x, partial f/partial y
Main idea	Change y as x changes	Change output with one variable while others are fixed
Example	y=x^2 -> dy/dx=2x	f=x^2+y^2 -> f_x=2x, f_y=2y

7. More Quick Examples

Function	Derivative / Partial derivative
y = 7x^4	dy/dx = 28x^3
y = 5x^3 + 4x	dy/dx = 15x^2 + 4
f(x,y)=xy	f_x = y, f_y = x
f(x,y)=5xy	f_x = 5y, f_y = 5x
f(x,y)=x^2y^2	f_x = 2xy^2, f_y = 2x^2y
f(x,y)=3x^2+4y^2	f_x = 6x, f_y = 8y
f(x,y)=x^3+2xy+5y^2	f_x = 3x^2+2y, f_y = 2x+10y

8. Practice Questions

1. Differentiate y = 6x^4 + 3x^2 - 8x + 10.

2. Differentiate y = (x^2 + 4)^2 using the chain rule.

3. Find partial f/partial x and partial f/partial y for f(x,y) = x^3 + y^3.

4. Find both partial derivatives for f(x,y) = 4x^2y + 3xy^2.

5. Find partial derivatives with respect to x, y and z for f(x,y,z) = xyz + x^2 + z^2.

Answers

1. 24x^3 + 6x - 8

2. 4x(x^2 + 4)

3. f_x = 3x^2; f_y = 3y^2

4. f_x = 8xy + 3y^2; f_y = 4x^2 + 6xy

5. f_x = yz + 2x; f_y = xz; f_z = xy + 2z

> **Remember:** In an ordinary derivative, differentiate with respect to the main independent variable. In a partial derivative, choose one variable and temporarily treat every other independent variable as a constant.